

School:Campus:

Academic Year: Subject Name: Subject Code:

Semester: Program: Branch: Specialization:

Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment: Build a Market – Basic NFT Marketplace Logic

***Coding Phase: Pseudo Code / Flow Chart / Algorithm**

1. Open Remix IDE and create two smart contracts, then compile them.
 - Build a SimpleNFT contract that allows users to mint NFTs using a token URI.
 - Build an NFTMarketplace contract that enables users to list their NFTs for sale and allows others to purchase them.
2. Deploy both contracts by selecting Injected Provider - MetaMask as the environment.
 - Deploy the SimpleNFT contract to the blockchain and copy its deployed address.
 - Deploy the NFTMarketplace contract to the blockchain and copy its deployed address.
3. Set up the frontend by creating a React web application. Add both contract addresses and their ABI files to the project, and use ethers.js to connect and interact with the blockchain.
4. Prompt the user to connect their crypto wallet, such as MetaMask.
5. Allow the user to mint an NFT by entering a Token URI (the link to NFT metadata hosted on Pinata).
6. After minting, retrieve and display the Token ID of the newly created NFT.
7. Enable users to list their NFTs for sale by entering the NFT contract address, Token ID, and price in ETH.
8. Approve the marketplace contract to manage and handle transactions for that specific NFT.
9. Call the list function on the marketplace contract using the provided details. The marketplace will generate a unique Listing ID for the sale.
10. Allow buyers to purchase NFTs by entering the Listing ID of the NFT they wish to buy.
11. Call the buy function on the marketplace contract, sending the required ETH amount. Once confirmed, ownership of the NFT will transfer from the seller to the buyer.

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*** As applicable according to the experiment.
Two sheets per experiment (10-20) to be used.**

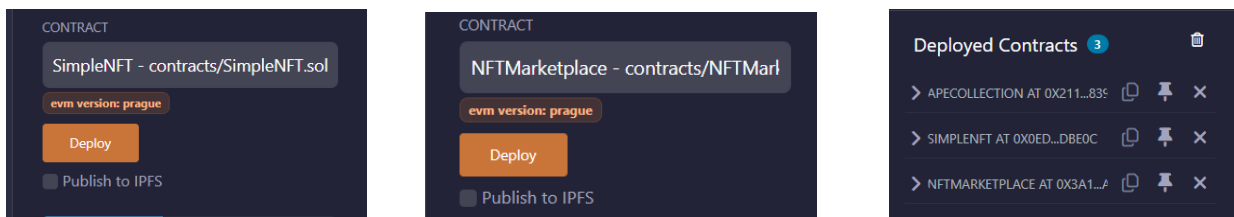
* Software used:

- Web browser
- Meta Mask wallet
- Sepolia Testnet
- Remix IDE

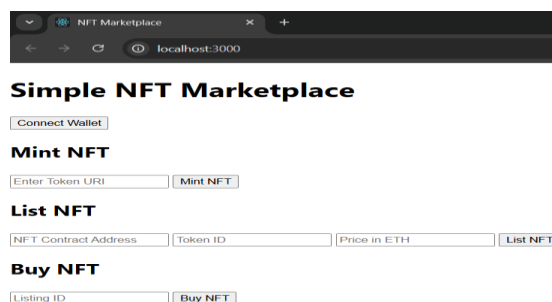
* Testing Phase: Compilation of Code (error detection)

No error

* Implementation Phase: Final Output (no error)



```
frontend > src > JS App.js > | nftAddress
1 import React, { useState } from "react";
2 import { ethers } from "ethers";
3
4 let signer;
5 let nftContract;
6 let marketplaceContract;
7
8 const nftAddress = "0x0Edad68312c2b3E21384D706a2B3bB363d00be0d";
9 const marketplaceAddress = "0x3a1d68e750b55f74586c6B4ECCd0c9d014Ea4FD6";
10
11 function App() {
12   const [walletAddress, setWalletAddress] = useState("");
13
14   async function connectWallet() {
15     if (window.ethereum) {
16       await window.ethereum.request({ method: "eth_requestAccounts" });
17
18       const provider = new ethers.providers.Web3Provider(window.ethereum); // Ethers v
19       signer = provider.getSigner();
20
21       const address = await signer.getAddress();
22       setWalletAddress(address);
23
24       const nftAbi = await (await fetch("./abi/SimpleNFT.json")).json();
25       const marketAbi = await (await fetch("./abi/NFTMarketplace.json")).json();
```



Simple NFT Marketplace

Connect Wallet

Connected: 0xc006C5491d961A76180582D1967D05C104f016C7

localhost:3000 says

Wallet connected and contracts loaded

OK

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* Implementation Phase: Final Output (no error)

Simple NFT Marketplace

Connect Wallet

Connected: 0xc006C5491d961A76180582D1967D05C104f016C7

Mint NFT

<https://tan-tricky-basilisk-33/> Mint NFT

localhost:3000 says

NFT Minted!

OK

Simple NFT Marketplace

Connect Wallet

Connected: 0xc006C5491d961A76180582D1967D05C104f016C7

Mint NFT

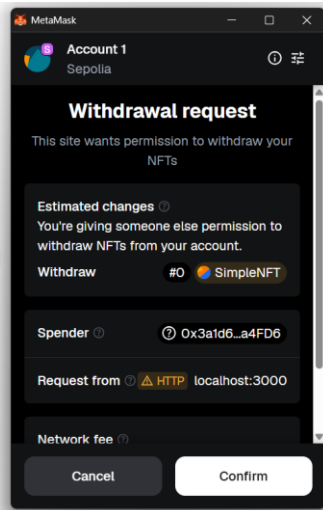
<https://tan-tricky-basilisk-33/> Mint NFT

List NFT

List NFT

Buy NFT

Buy NFT



localhost:3000 says

NFT Listed!

OK

Simple NFT Marketplace

Connect Wallet

Connected: 0xc006C5491d961A76180582D1967D05C104f016C7

Mint NFT

<https://tan-tricky-basilisk-33/> Mint NFT

List NFT

List NFT

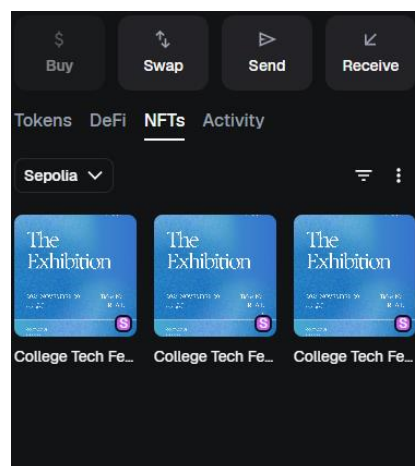
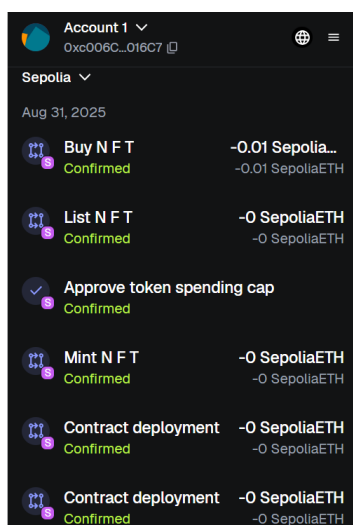
Buy NFT

Buy NFT

localhost:3000 says

NFT Purchased!

OK



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* Observation :

- From this experiment we observed how to deploy NFT and marketplace contracts.
- Connect a crypto wallet to the app.
- Mint NFTs by providing metadata URI and retrieve Token IDs after minting.
- Approve and list NFTs for sale with prices.
- Use Listing IDs to buy NFTs securely.
- Transfer ownership and payment on purchase.
- Build a simple UI to interact with blockchain.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student :

Name :

Regn. No. :

Signature of the Faculty :

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